

Lab 7: Generate and Implement On-Page SEO Meta

Mapping: LO3 | A2 | K10 Objective: Use prompts to create and implement accurate on-page elements Deliverable: Optimised static product page and metadata decision log Primary source: <https://developers.google.com/search/docs/appearance/title-link>

Scenario

Titles and snippets should accurately represent visible content. Structured data must match the page and does not guarantee a rich result.

The EcoDesk product page is useful but generic. The team needs stronger search clarity without inventing prices, ratings, stock or environmental claims.

Training assumptions and guardrails

- Product names and descriptions are fictional.
- No reviews, aggregate rating or certification markup may be added.
- The canonical URL is a training placeholder and must be changed before real deployment.
- Title and description lengths are editorial guidance, not ranking rules.

Questions

1. Which page facts can safely appear in metadata?
2. Which generated option best matches intent and brand?
3. Which structured-data type is eligible and visible?
4. Which internal links reduce user effort?

Detailed step-by-step procedure

Step 1: Extract page facts

Create a fact table from visible product content and approved business information.

Checkpoint: Save the evidence, file or decision produced before continuing.

Step 2: Generate options

Run Template 1 (single page) from the prompt pack for five titles and descriptions with rationale, distinct angle and prohibited-claim check, then Template 3 (local intent) to test whether a Singapore or trust signal is honestly available.

Checkpoint: Save the evidence, file or decision produced before continuing.

Step 3: Select and edit

Choose one option, improve natural language and confirm it matches the page.

Checkpoint: Save the evidence, file or decision produced before continuing.

Step 4: Implement head elements

Add title, description, canonical, robots, Open Graph and social-card metadata, verifying each length in a pixel-width checker rather than trusting the model count.

Checkpoint: Save the evidence, file or decision produced before continuing.

Step 5: Improve structure

Use one clear H1, descriptive sections, useful copy and accessible image attributes.

Checkpoint: Save the evidence, file or decision produced before continuing.

Step 6: Add internal links

Connect the page to the pillar, blog and enquiry path with descriptive anchors.

Checkpoint: Save the evidence, file or decision produced before continuing.

Step 7: Add JSON-LD

Use eligible visible product/organisation facts only and keep JSON valid.

Checkpoint: Save the evidence, file or decision produced before continuing.

Step 8: Validate

Inspect source, HTML, mobile layout and structured data; record limitations.

Checkpoint: Save the evidence, file or decision produced before continuing.

Use the shared EcoDesk website

Live site: <https://tertiarycourses.github.io/TGS-2020503501-Generative-AI-for-Search-Engine-Optimization-SEO/>

1. Keep this activity folder open for its prompts and evidence files.
2. From the repository root, run: `python3 -m http.server 8000 --directory activities/ecodesk-demo-website`
3. Open `http://localhost:8000/` or use the live site above.
4. Apply this activity to the same shared website used in every other activity.
5. Test at desktop and mobile widths.

Acceptance criteria

- [] Metadata is unique and accurate, visible content and JSON-LD agree, links work, images are accessible and no unsupported rating, price or guarantee is present.
- [] All links, forms, assistant and mobile navigation work.
- [] Work is applied to the one shared EcoDesk website, not a separate copy.
- [] AI output is reviewed and the decision log is complete.